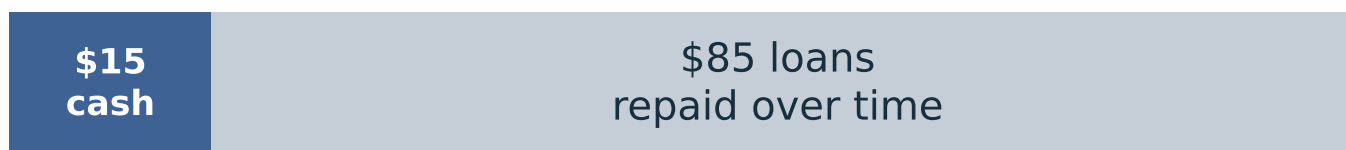


Imagine a bank with \$100 of assets.

It has \$15 in cash. Borrowers owe it \$85 in loans, which they will repay over time.



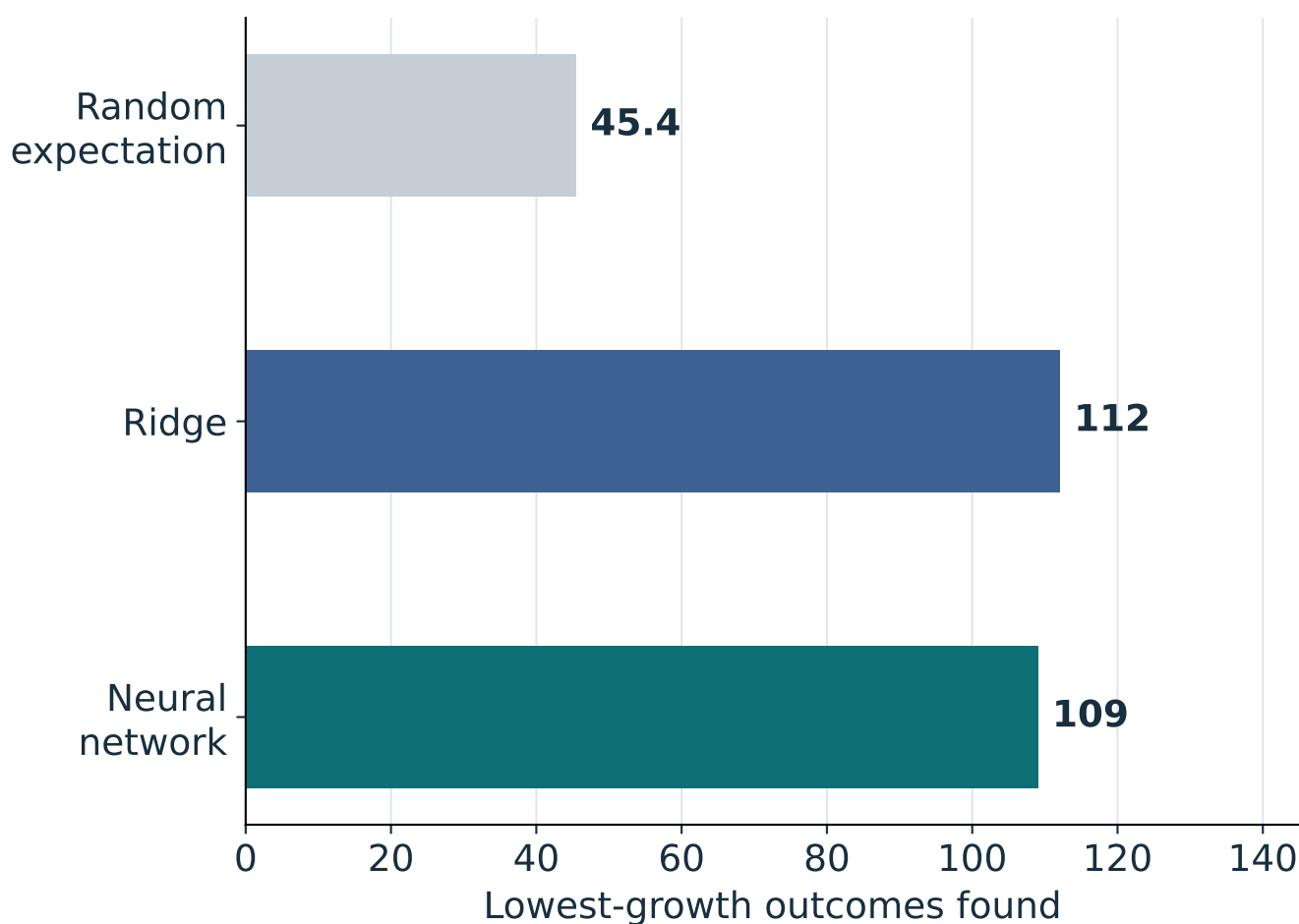
Now customers ask to withdraw \$20.



Customers request \$20 today. The bank needs another \$5 in cash. It could borrow or sell an asset. The next notebook step explains why equity is separate from available cash.

4,536 banks. 454 places to look.

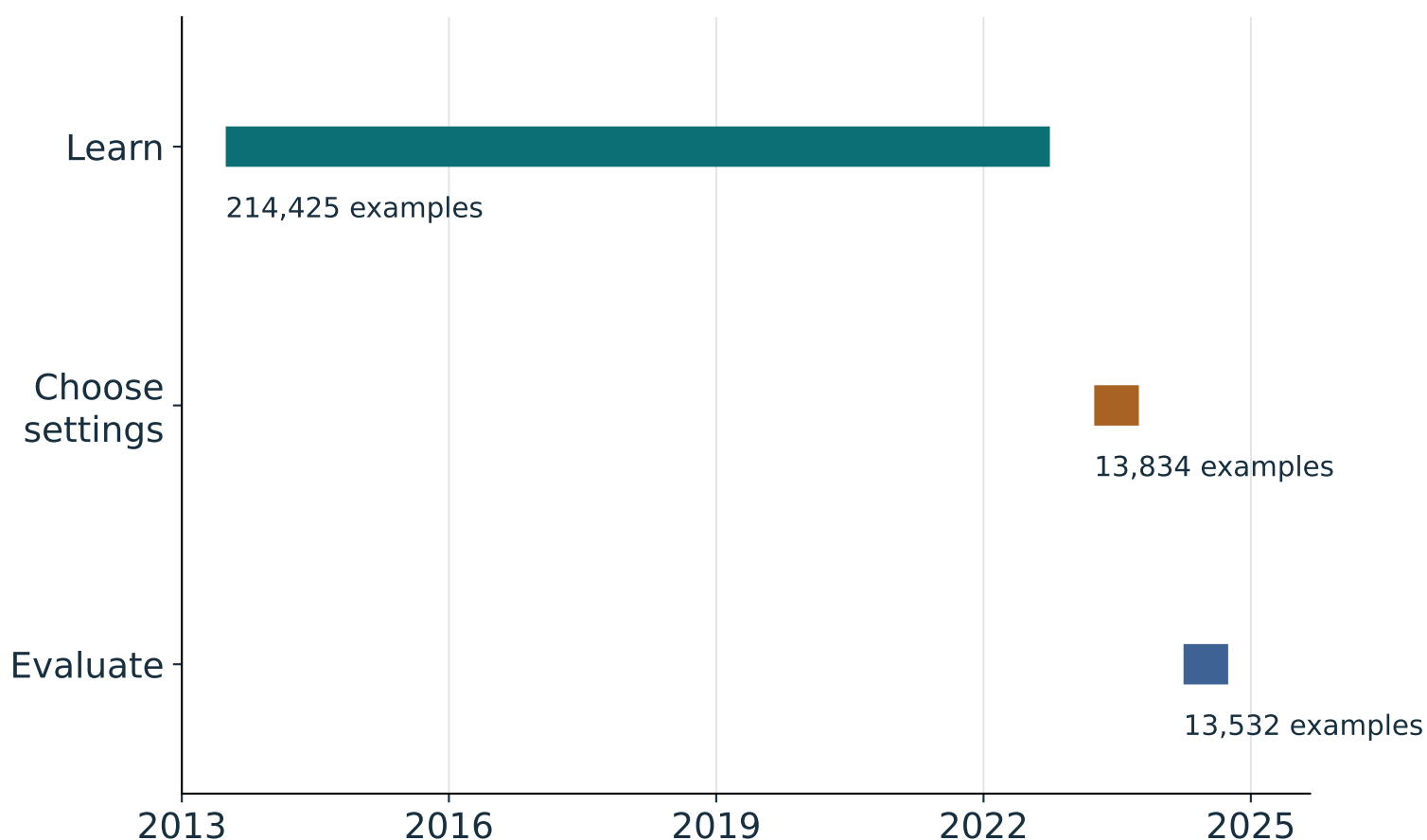
Can bank reports help us decide which banks to examine first? March 2024 reports predict June outcomes.



The experiment assumes room to examine 10% of banks. These counts show selected banks later in the lowest-growth 10%. A person would investigate the reason for the movement.

Twelve years. Three different jobs.

48 source quarters from 2013–2024. Training uses 38 predictor quarters; validation and evaluation use three each.



2024 has already been examined during development. Thousands of bank examples share three evaluation dates. A later untouched period would test performance under new conditions.

Five inputs describe the bank today.

Each prediction uses current balance-sheet information and the previous quarter's deposit change.

Deposit size

Log of domestic deposits

Cash / assets

Cash and balances due, relative to assets

Loans / assets

Net loans and leases, relative to assets

Equity / assets

Capitalization

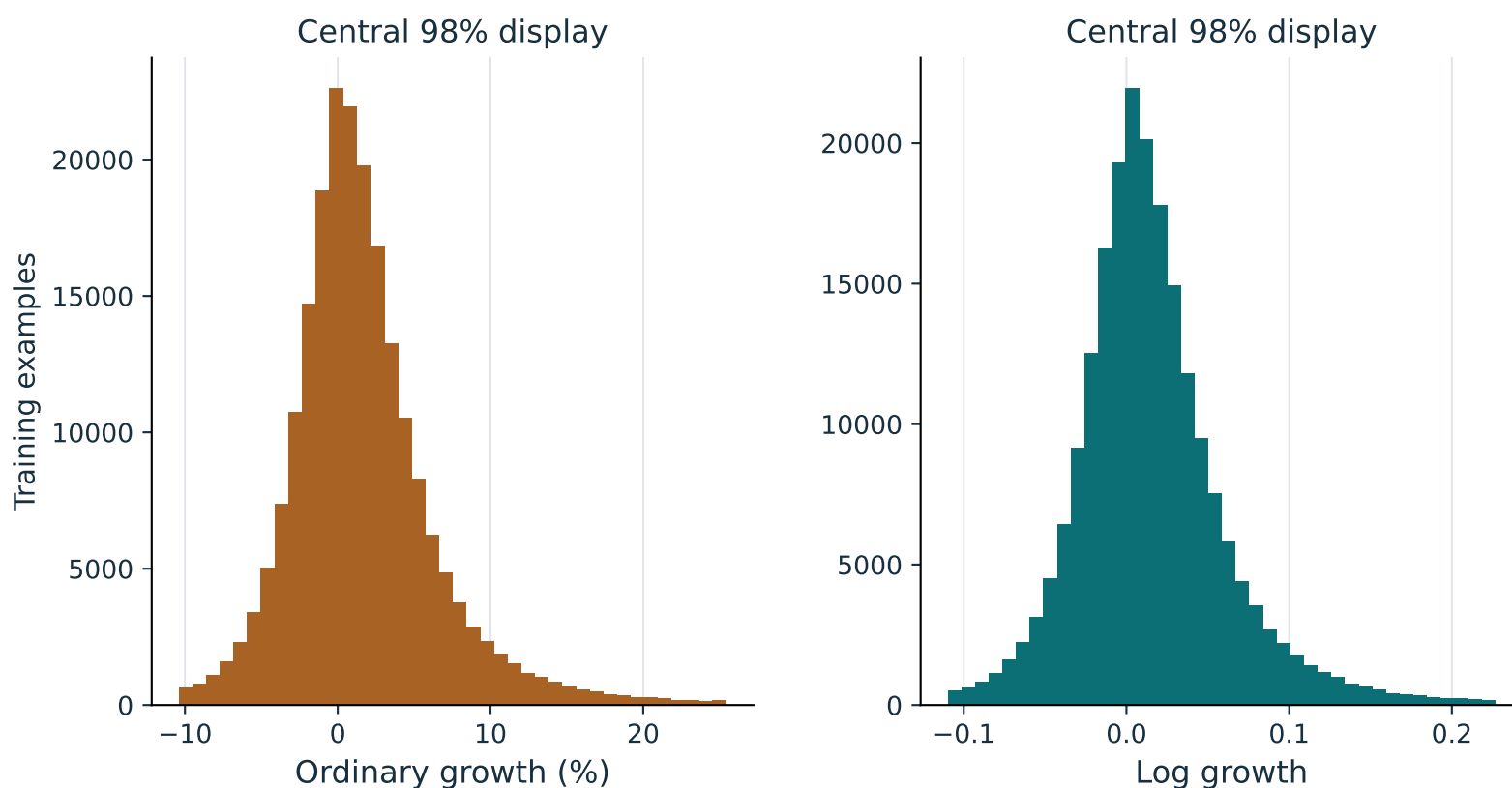
Previous growth

Prior-quarter log deposit growth

The output is next-quarter deposit growth. These five inputs give a partial view of a bank; a human would examine funding composition and institutional context too.

A tiny starting balance makes a huge percentage.

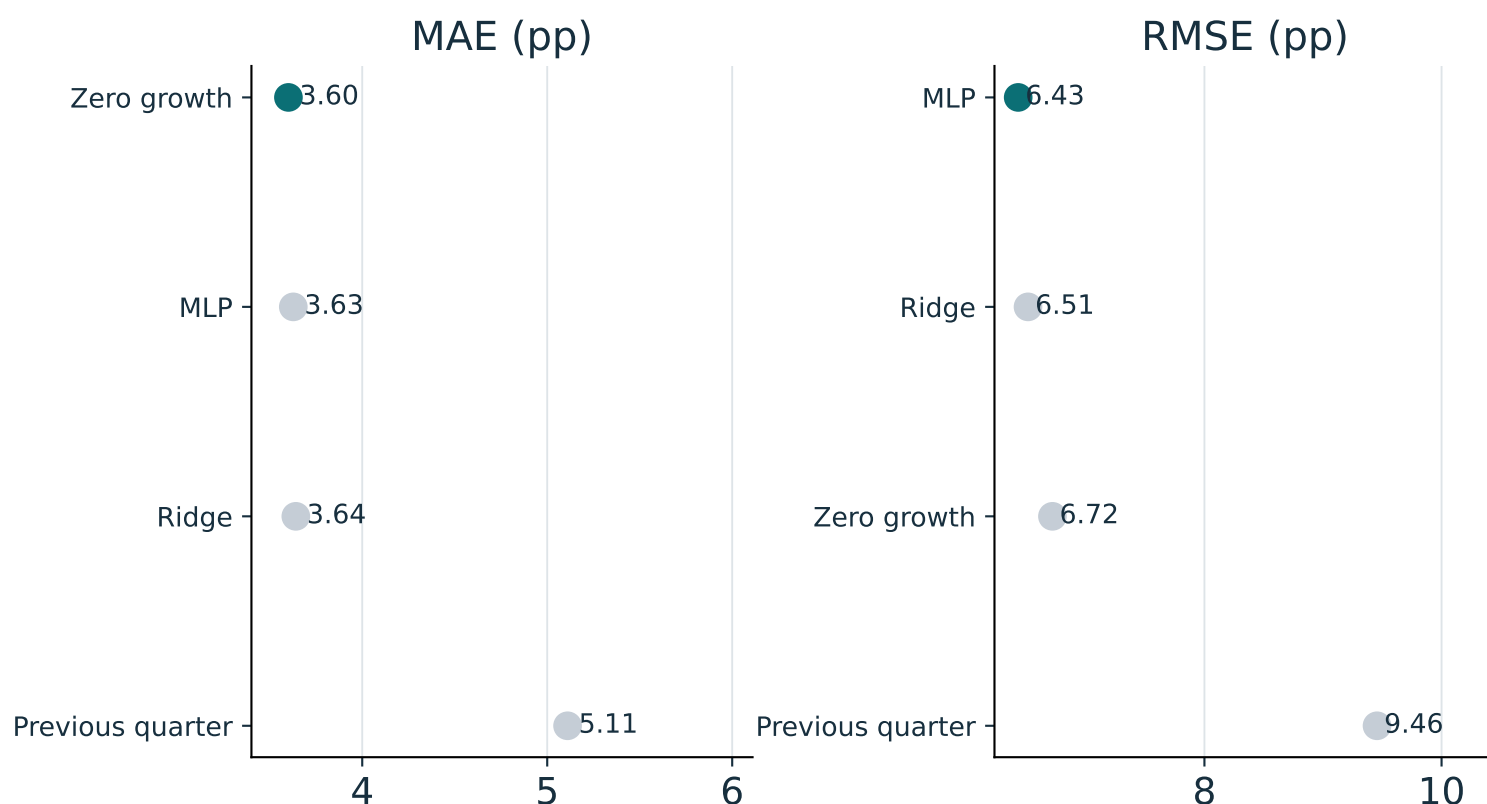
Training example: \$1.15 million → \$6.01 billion. The reported increase is about 522,646%; log growth is about 8.56.



Both panels use the same training population and show their own central 98% windows. Extreme valid observations stay in training. The economic cause of this example remains unverified.

The error measure changes the leader.

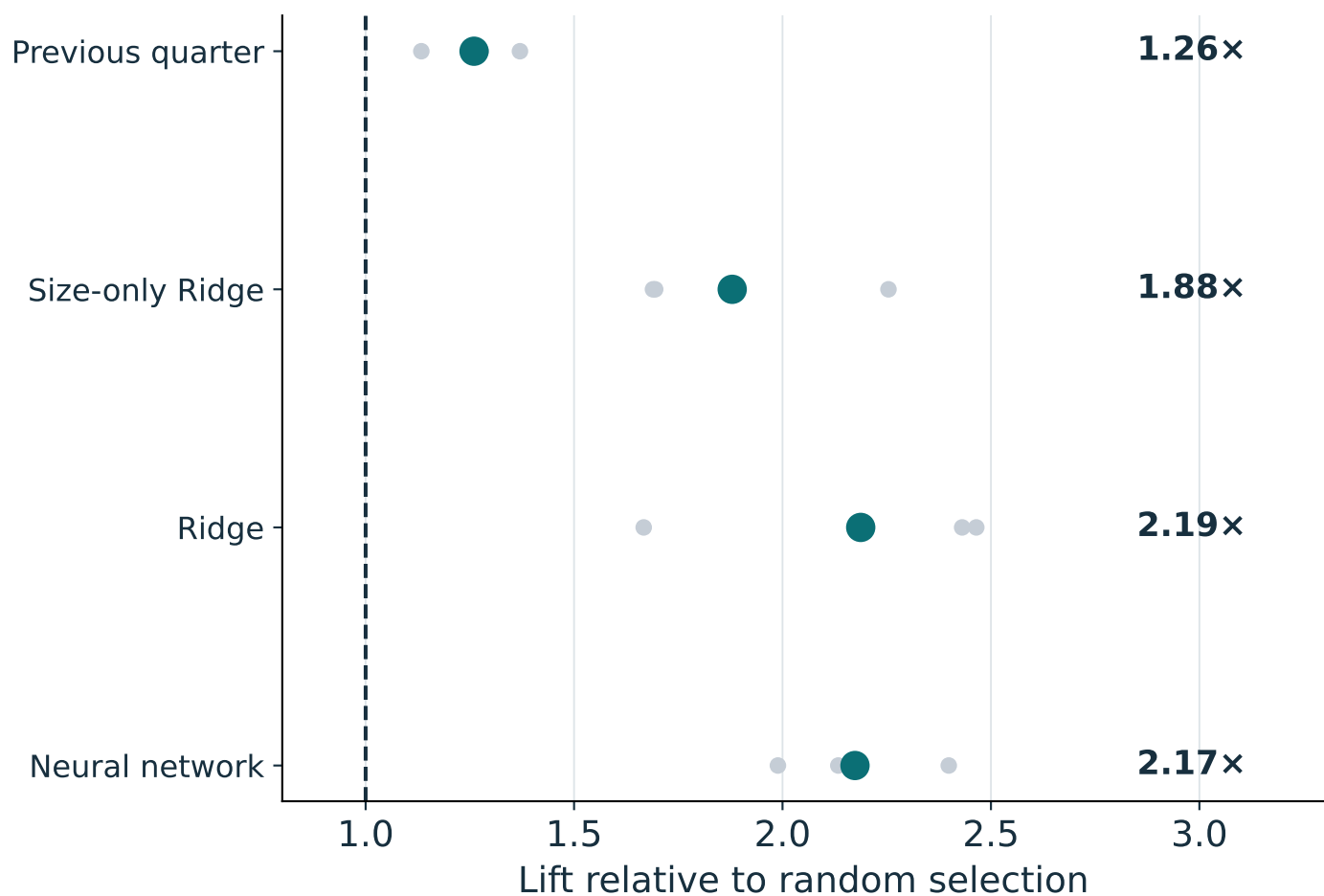
Original four-model comparison. All models are scored on the same 13,532 reused-2024 bank-quarter examples.



Zero growth has the lowest mean absolute error. The neural network has the lowest RMSE, where large misses receive more weight. Follow-up baselines appear later.

About twice the concentration of low-growth outcomes.

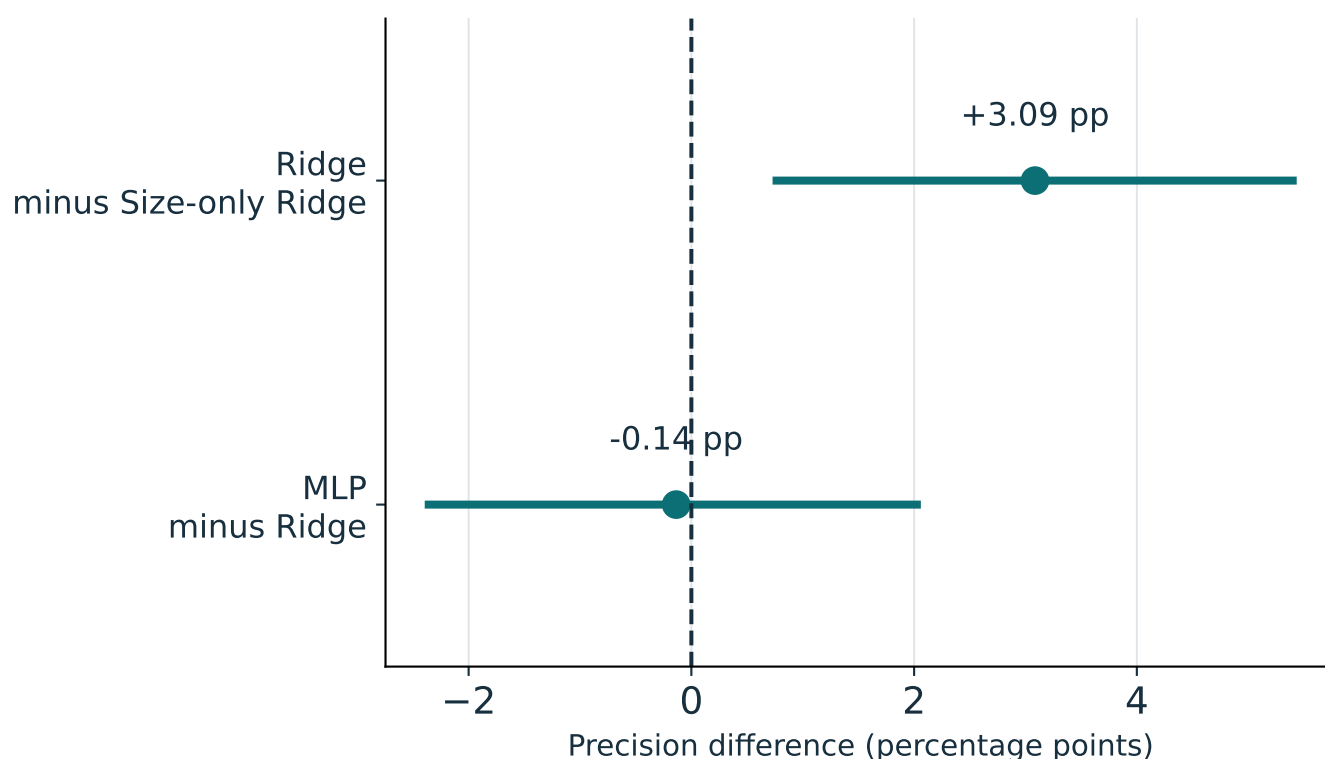
Select the lowest predicted-growth 10% each quarter. Lift compares the share found with the exact random expectation.



Large dots show equal-quarter averages; small dots show individual quarters. Size-only Ridge is a follow-up benchmark developed after viewing 2024. March alone has different results from this average.

The MLP-Ridge interval includes either ordering.

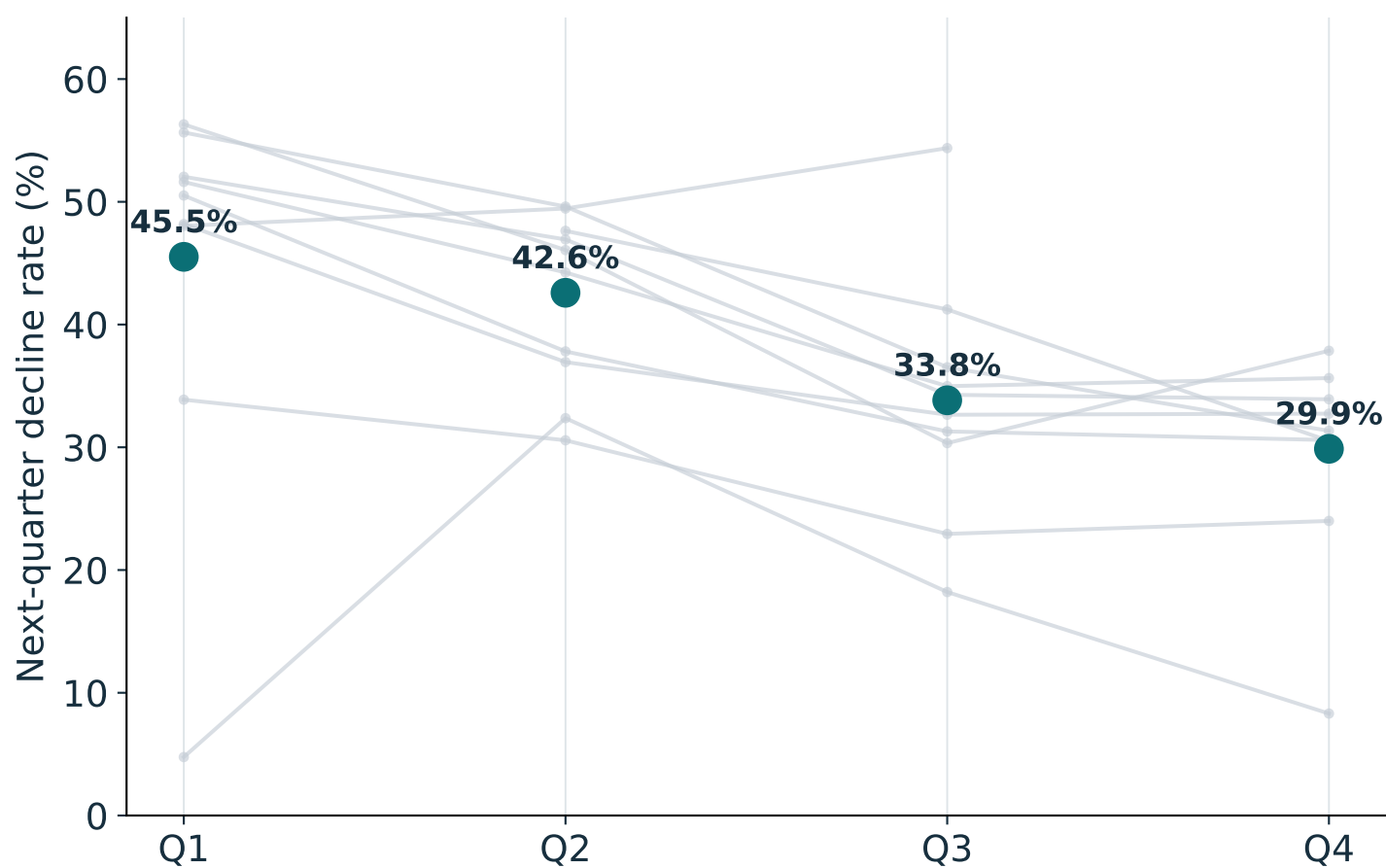
Resample banks 1,000 times, keeping each bank's repeated quarters together. Keep the fitted models and review lists fixed.



A range crossing zero includes either ordering. These intervals describe variation across the sampled banks in the same three quarters. They do not measure uncertainty across future economic periods.

A seasonal baseline improves historical MAE.

Training data only. Each faint line is a year. Large dots pool bank-quarter decline rates within each predictor quarter.



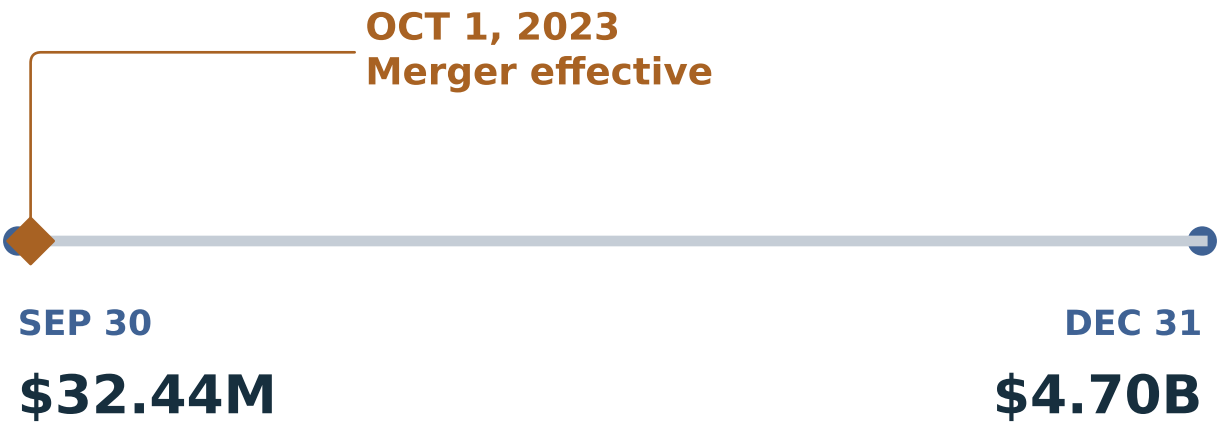
The seasonal forecast uses training-period median growth for that quarter of the year. Its reused-2024 score is a follow-up result; the rule needs a later untouched evaluation.

A merger falls inside the forecast interval.

Plus International Bank, CERT 57083. Emigrant merged into Plus under the Emigrant name on October 1, 2023.

+14,380%

Reported deposit growth

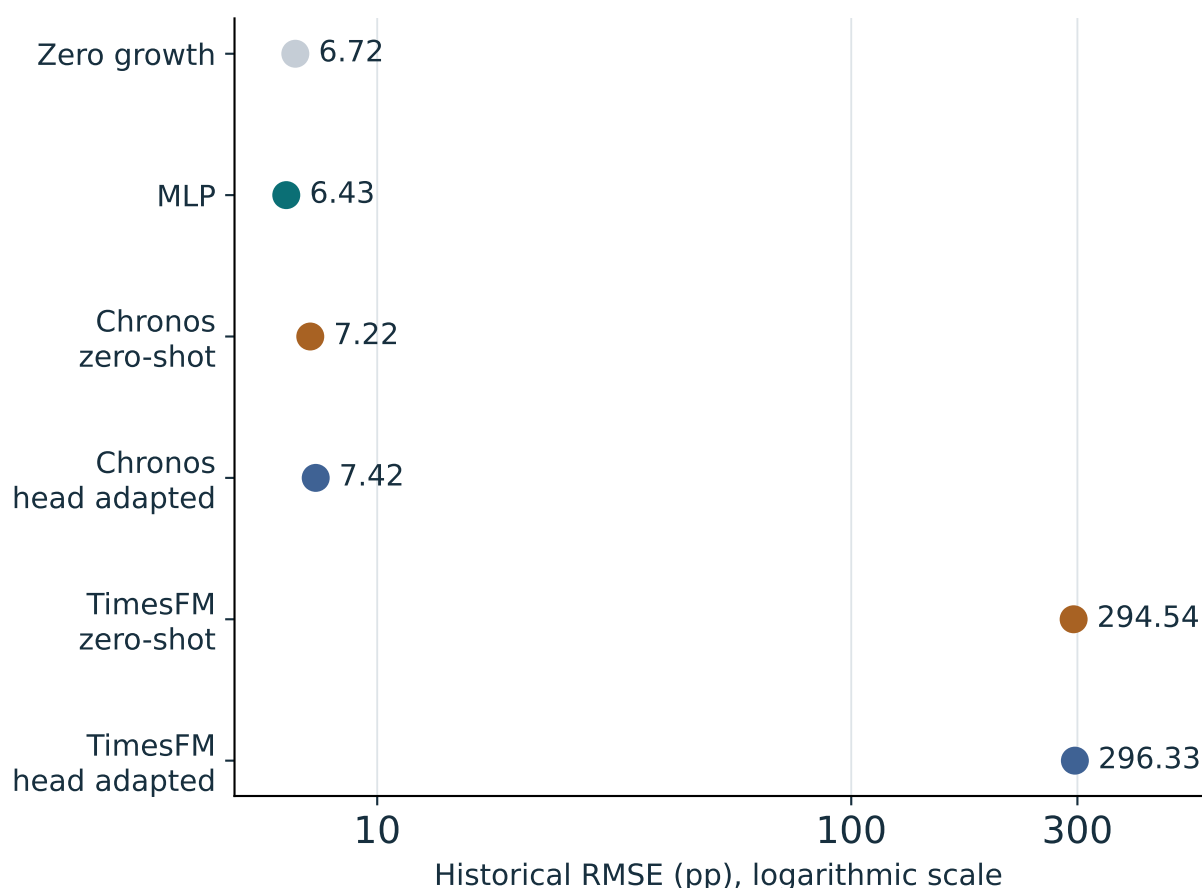


Actual calendar spacing. A merger falls inside the interval.

The two-case event list is incomplete. Primary scores retain all eligible observations. The date source was published in November 2024 and supports retrospective annotation.

Pretraining and adaptation still need a benchmark.

TimesFM and Chronos were tested with frozen weights, then with their forecast heads adapted using earlier data.



One bounded head-adaptation recipe was selected using validation. Neither adapted model beat the original baselines. TimesFM's large error motivates the matching-history checks in the notebook.

Freeze the rules. Test the next unseen quarter.

Use the experiment to decide which approaches deserve a stronger test.

Keep simple forecasts in the comparison.

Seasonal median has 3.500 pp MAE in the expanded historical comparison.

Test whether the review list helps.

Ridge averaged 2.19× random concentration. Size alone averaged 1.88×.

Investigate the underlying bank.

Check funding, deposit composition, mergers, and institution history before drawing conclusions.

Freeze the candidate models, event rules, and review capacity. Evaluate a later period that has not guided development. Human investigation remains essential; no operational savings were measured.